

Payment & Financial Services - Document 4 - Page 1/6

Document ID: PAY-004 | Classification: Internal Use

Payment processing performance monitoring: latency metrics, throughput, success rates, and SLA thresholds.

Webhook retry: 3 attempts at 1h, 4h, 24h. Implement idempotent handlers for duplicate events.

Payment processor integration: Stripe, Adyen, Braintree API documentation and migration guides.

Key considerations for payment systems: secure tokenization, PCI-DSS compliance, audit logging. For EU transactions ensure PSD2/SCA. For USD/INR verify FX rates and conversion timestamps.

Troubleshooting:

- Check gateway health endpoint. Verify SSL certificate expiry.
- Review transaction logs for error codes. 5003 = timeout, 4001 = invalid method.
- For currency mismatch: verify amount in minor units, currency ISO code.

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Payment database management: transaction logs, audit trails, backup procedures, and data retention policies.

Gateway failover: primary EU, secondary US. Health check every 30s. Auto-switch on 3 consecutive failures.

Financial compliance: PCI-DSS Level 1, SOC 2, ISO 27001 requirements for payment systems.

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Payment gateway configuration for EU region: PSD2 compliance, SCA requirements, and GDPR data handling.

Refund must be $\leq$ original amount. Partial refunds supported. Full refund voids original transaction.

Transaction reconciliation: daily settlement, mismatch resolution, and reconciliation report generation.

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Payment user authentication: 3DS2 flows, biometric verification, and risk-based authentication.

3DS2 challenge: customer may receive OTP from bank. Timeout 5 minutes. Handle challenge completion.

Payment fraud detection: velocity checks, device fingerprinting, and ML-based risk scoring.

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Payment transaction logging: structured logs, trace IDs, and observability for incident investigation.

Settlement cycle: T+2 for cards, T+1 for bank transfer. Check processor-specific timelines.

Merchant account management: onboarding, KYC verification, and settlement schedule configuration.

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Payment service deployment: blue-green releases, canary deployments, and rollback procedures.

Payment reconciliation: match transaction ID, amount, currency. Flag mismatches for manual review.

Payment webhooks: event types, signature verification, retry policy, and idempotency handling.

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